

DBMS — LECTURE 1: INTRO TO DATABASE, DBMS & RDBMS 📝✨

1. What is a Database System?

CORE FORMULA:

Database System = Database + DBMS

- **Database:** A collection of **related data** stored in an organized manner.
- **DBMS:** Software that allows users to **store, access, insert, delete, and update** data.



Real Example (IRCTC Railway System):

Train Info | Passenger Info | Reservation | Tickets | Stations
All stored in DBs, managed by DBMS for seamless access!

2. What is Data?

Raw facts or information that can be stored & processed.

- Student Name → **Husen**
- Age → **20** | Course → **BCA** | Marks → **85**

Collected related data = Useful Information!

3. What is a Database?

Collection of RELATED data (Keyword: **RELATED**!).

University DB:

Students ↔ Faculty ↔ Courses ↔ Marks ↔ Attendance

📌 **EXAM TIP / IMPORTANT POINT:** DB is NOT random data! Passport data doesn't belong in IRCTC DB!

4. Types of Data

DATA → **Structured** | **Semi-Structured** | **Unstructured**

5. Structured Data

- Follows a **predefined structure/format** (commonly **Tables**).
- Columns = defined meaning | Rows = individual records.
- *Examples:* University, Railway, Banking, Employee, Customer records.

Student_ID	Name	Course	Marks
101	Ali	BCA	85
102	Ravi	BCA	78
103	Sara	BCA	92

6. Unstructured Data

- **No predefined tabular structure.** (*Not useless!* Just doesn't fit tables).
- *Examples:* Images, Videos, Audio, Social Media, Webpages, Chats.

7. Semi-Structured Data

- No rigid table structure, but contains organizational tags/keys/metadata.
- *Examples:* JSON, XML, HTML

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JSON Ex: { "name": "Husen", "course": "BCA", "marks": 85 }
```

8. Structured vs Unstructured Data

Feature	Structured Data	Unstructured Data
Structure	Predefined	No fixed structure
Common Format	Tables	Files / Media / Docs
Easy SQL Query?	YES	Generally NOT directly
Technology	RDBMS (MySQL, PostgreSQL, SQL Server)	Big Data / NoSQL (Hadoop, Data Lakes)

9 & 10. What is DBMS & Why Do We Need It?

DBMS = Database Management System. Provides interface to store, manage, access & manipulate data efficiently.

11. CRUD Operations ⚡ (Super Important!)

- **C**reate → Add new data (e.g. Add passenger)
- **R**ead → Retrieve/access data (e.g. Find trains from BLR to DEL)
- **U**ppdate → Modify existing data (e.g. Change address)
- **D**elete → Remove data (e.g. Cancel reservation)

12. DBMS Hides Physical Storage (Data Abstraction)

User → DBMS → Database → Physical Storage (e.g. Hard Disk in Mumbai)
User doesn't need to know physical server/disk details! DBMS handles it.

13. Popular DBMS Software

- **Microsoft SQL Server** (Microsoft) | **Oracle Database** (Oracle)
- **MySQL** (Oracle) | **IBM Db2** (IBM) | PostgreSQL, SQLite, MariaDB

14, 15 & 16. What is RDBMS & Technical Terms ★

RDBMS = Relational Database Management System (based on the **Relational Model** using Tables/Relations).

 INTERVIEW & EXAM TERMINOLOGY TRICK:

Common Term	Relational / Technical Term
Table	Relation
Row	Tuple
Column	Attribute
Cell Value	Attribute Value

17 & 18. Why RDBMS & DBMS vs RDBMS

- RDBMS provides structured tables, rows, columns, constraints, and **relationships between tables using Keys**.
- **Every RDBMS is a DBMS, but DBMS is a broader term!**

19 & 20. Architecture & Real-World Example

RDBMS Schema Example:

[STUDENT] —(relates to)—> [COURSE] —(relates to)—> [DEPARTMENT]

IRCTC DB Tables: Train_Table + Passenger_Table + Reservation_Table (connected via Keys)

23. Most Important Exam Points ★ (Memorize!)

- ✓ **Database System** = Database + DBMS
- ✓ **Relation** = Table | **Tuple** = Row | **Attribute** = Column
- ✓ **Structured** = Predefined table structure | **Unstructured** = No fixed table structure
- ✓ **CRUD** = Create, Read, Update, Delete
- ✓ **RDBMS Systems:** MySQL, Oracle, SQL Server, PostgreSQL, Db2

24. Quick Revision Interview Questions 🎯

1. **What is a DB?** Organized collection of related data.
2. **DBMS vs RDBMS?** DBMS is general software; RDBMS specifically uses the relational model (tables & keys).
3. **What is a Tuple & Attribute?** Tuple = Row, Attribute = Column in a Relation.
4. **What are CRUD operations?** Create, Read, Update, Delete.

 **END OF LECTURE 1 NOTES — ALL THE BEST FOR EXAMS! 100**